

■ 복제 데이터베이스

KEY WORDS

- Clone DB 생성(CP,
- Rman 을 이용한 Oracle DB 복제(Active Duplication)

<<시나리오 1>> CLONE DB 생성하기

■ 복제 데이터베이스

- Host 에서

1. 백업 데이터파일을 clone 디렉터리에 복사

```
[oracle@oel7v9r2 ~]$ mkdir clone
[oracle@oel7v9r2 ~]$ cp -v backup/open_bkp/*.dbf clone/
'backup/open_bkp/sysaux01.dbf' -> 'clone/sysaux01.dbf'
'backup/open_bkp/system01.dbf' -> 'clone/system01.dbf'
'backup/open_bkp/temp01.dbf' -> 'clone/temp01.dbf'
'backup/open_bkp/undotbs01.dbf' -> 'clone/undotbs01.dbf'
'backup/open_bkp/users01.dbf' -> 'clone/users01.dbf'
```

2. archive log file 을 clone 디렉터리에 복사

```
[oracle@oel7v9r2 ~]$ cp -v arch1/* clone/
'arch1/0001_0000000001_1201184496.arc' -> 'clone/0001_0000000001_1201184496.arc'
'arch1/0001_0000000002_1201184496.arc' -> 'clone/0001_0000000002_1201184496.arc'
'arch1/0001_0000000003_1201184496.arc' -> 'clone/0001_0000000003_1201184496.arc'
'arch1/0001_0000000004_1201184496.arc' -> 'clone/0001_0000000004_1201184496.arc'
'arch1/0001_0000000005_1201184496.arc' -> 'clone/0001_0000000005_1201184496.arc'
'arch1/0001_0000000006_1201184496.arc' -> 'clone/0001_0000000006_1201184496.arc'
'arch1/0001_0000000024_1200585133.arc' -> 'clone/0001_0000000024_1200585133.arc'
'arch1/0001_0000000025_1200585133.arc' -> 'clone/0001_0000000025_1200585133.arc'
'arch1/0001_0000000026_1200585133.arc' -> 'clone/0001_0000000026_1200585133.arc'
'arch1/0001_0000000027_1200585133.arc' -> 'clone/0001_0000000027_1200585133.arc'
'arch1/0001_0000000028_1200585133.arc' -> 'clone/0001_0000000028_1200585133.arc'
'arch1/0001_0000000029_1200585133.arc' -> 'clone/0001_0000000029_1200585133.arc'
'arch1/0001_0000000030_1200585133.arc' -> 'clone/0001_0000000030_1200585133.arc'
'arch1/0001_0000000031_1200585133.arc' -> 'clone/0001_0000000031_1200585133.arc'
'arch1/0001_0000000032_1200585133.arc' -> 'clone/0001_0000000032_1200585133.arc'
'arch1/0001_0000000033_1200585133.arc' -> 'clone/0001_0000000033_1200585133.arc'
'arch1/0001_0000000034_1200585133.arc' -> 'clone/0001_0000000034_1200585133.arc'
```

```
'arch1/0001_0000000035_1200585133.arc' -> 'clone/0001_0000000035_1200585133.arc'
'arch1/0001_0000000036_1200585133.arc' -> 'clone/0001_0000000036_1200585133.arc'
'arch1/0001_0000000037_1200585133.arc' -> 'clone/0001_0000000037_1200585133.arc'
'arch1/0001_0000000038_1200585133.arc' -> 'clone/0001_0000000038_1200585133.arc'
'arch1/0001_0000000040_1200585133.arc' -> 'clone/0001_0000000040_1200585133.arc'
'arch1/0001_0000000041_1200585133.arc' -> 'clone/0001_0000000041_1200585133.arc'
'arch1/0001_0000000042_1200585133.arc' -> 'clone/0001_0000000042_1200585133.arc'
'arch1/0001_0000000043_1200585133.arc' -> 'clone/0001_0000000043_1200585133.arc'
'arch1/0001_0000000044_1200585133.arc' -> 'clone/0001_0000000044_1200585133.arc'
'arch1/0001_0000000045_1200585133.arc' -> 'clone/0001_0000000045_1200585133.arc'
'arch1/0001_0000000046_1200585133.arc' -> 'clone/0001_0000000046_1200585133.arc'
'arch1/0001_0000000047_1200585133.arc' -> 'clone/0001_0000000047_1200585133.arc'
'arch1/0001_0000000048_1200585133.arc' -> 'clone/0001_0000000048_1200585133.arc'
'arch1/0001_0000000049_1200585133.arc' -> 'clone/0001_0000000049_1200585133.arc'
'arch1/0001_0000000050_1200585133.arc' -> 'clone/0001_0000000050_1200585133.arc'
'arch1/0001_0000000051_1200585133.arc' -> 'clone/0001_0000000051_1200585133.arc'
```

DB 에서

3. 오라클의 pfile 생성

```
SQL> create pfile='/home/oracle/clone/initclone.ora' from spfile;
```

File created.

```
SQL> ! ls /home/oracle/clone
```

```
0001_0000000001_1201184496.arc 0001_0000000029_1200585133.arc 0001_0000000041_1200585133.arc contr
ol01.ctl
0001_0000000002_1201184496.arc 0001_0000000030_1200585133.arc 0001_0000000042_1200585133.arc defts
01.dbf
0001_0000000003_1201184496.arc 0001_0000000031_1200585133.arc 0001_0000000043_1200585133.arc initclo
ne.ora
0001_0000000004_1201184496.arc 0001_0000000032_1200585133.arc 0001_0000000044_1200585133.arc sysau
x01.dbf
0001_0000000005_1201184496.arc 0001_0000000033_1200585133.arc 0001_0000000045_1200585133.arc syste
m01.dbf
0001_0000000006_1201184496.arc 0001_0000000034_1200585133.arc 0001_0000000046_1200585133.arc temp
01.dbf
0001_0000000024_1200585133.arc 0001_0000000035_1200585133.arc 0001_0000000047_1200585133.arc undot
bs01.dbf
0001_0000000025_1200585133.arc 0001_0000000036_1200585133.arc 0001_0000000048_1200585133.arc users
01.dbf
0001_0000000026_1200585133.arc 0001_0000000037_1200585133.arc 0001_0000000049_1200585133.arc
```

```
0001_00000000027_1200585133.arc 0001_00000000038_1200585133.arc 0001_00000000050_1200585133.arc
0001_00000000028_1200585133.arc 0001_00000000040_1200585133.arc 0001_00000000051_1200585133.arc
```

4. control trace file 생성

```
SQL> alter database backup controlfile to trace as '/home/oracle/clone/control_clone.txt';
```

Database altered.

5. pfile 수정

```
[oracle@oracle19c clone]$ vi initclone.ora
```

```
*.compatible='19.0.0'
*.control_files='/home/oracle/clone/control01.ctl'
*.db_name='clone'
*.log_archive_dest_1='location=/home/oracle/clone mandatory'
*.log_archive_format='arch_%t_%s_%r.arc'
*.undo_tablespace='UNDOTBS1'
```

6. 새로운 control file 생성 - creclone.sql

```
STARTUP pfile=/home/oracle/clone/initclone.ora NOMOUNT
CREATE CONTROLFILE SET DATABASE "CLONE" RESETLOGS ARCHIVELOG
    MAXLOGFILES 16
    MAXLOGMEMBERS 3
    MAXDATAFILES 100
    MAXINSTANCES 8
    MAXLOGHISTORY 292
LOGFILE
    GROUP 1 '/home/oracle/clone/redo01.log' SIZE 20M BLOCKSIZE 512,
    GROUP 2 '/home/oracle/clone/redo02.log' SIZE 20M BLOCKSIZE 512,
    GROUP 3 '/home/oracle/clone/redo03.log' SIZE 20M BLOCKSIZE 512
DATAFILE
    '/home/oracle/clone/system01.dbf',
    '/home/oracle/clone/undotbs01.dbf',
    '/home/oracle/clone/sysaux01.dbf',
    '/home/oracle/clone/users01.dbf'
CHARACTER SET AL32UTF8
;
```

7. 복제 데이터베이스 생성

```
[oracle@oel7v9r2 ~]$ echo $ORACLE_HOME
```

```
/u01/app/oracle/product/19.3.0/dbhome_1
[oracle@oel7v9r2 ~]$ echo $ORACLE_SID
orcl
[oracle@oel7v9r2 ~]$ . oraenv
ORACLE_SID = [orcl] ? clone
ORACLE_HOME = [/home/oracle] ? /u01/app/oracle/product/19.3.0/dbhome_1
The Oracle base remains unchanged with value /u01/app/oracle
[oracle@oel7v9r2 ~]$ echo $ORACLE_SID
clone
```

```
[oracle@oel7v9r2 ~]$ sqlplus / as sysdba
Connected to an idle instance.
```

```
SQL> !vi creclone.sql
```

```
SQL> @creclone
```

Control file created.

```
SQL> recover database using backup controlfile until cancel
ORA-00279: change 2783381 generated at 05/15/2025 14:25:37 needed for thread 1
ORA-00289: suggestion : /home/oracle/clone/0001_0000000001_1201184496.arc
ORA-00280: change 2783381 for thread 1 is in sequence #1
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
auto
```

```
SQL> alter database open resetlogs;
Database altered.
```

```
SQL> ALTER TABLESPACE TEMP ADD TEMPFILE '/home/oracle/clone/temp01.dbf' REUSE;
Tablespace altered.
```

```
SQL> select name, created, log_mode from v$database;
```

NAME	CREATED	LOG_MODE
CLONE	15-MAY-25	ARCHIVELOG

```
SYS@clone> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS	ARC
FIRST_CHANGE#	NEXT_CHANGE#				
1	1	/home/oracle/clone/redo01.log	20		
CURRENT	NO	2783431	1.8447E+19		
2	0	/home/oracle/clone/redo02.log	20		
UNUSED	YES	0	0		
3	0	/home/oracle/clone/redo03.log	20		
UNUSED	YES	0	0		

```
SQL> !ls /home/oracle/clone
```

■ 복제 데이터베이스 삭제

```
SQL> shutdown abort
```

```
ORACLE instance shut down.
```

```
SQL> exit
```

```
[oracle@oel7v9r2 ~]$ cd $ORACLE_BASE/diag/rdbms
```

```
[oracle@oel7v9r2 ~]$ ls
```

```
clone orcl
```

```
[oracle@oel7v9r2 ~]$ rm -r clone/
```

```
[oracle@oel7v9r2 ~]$ ls
```

```
orcl
```

```
[oracle@oel7v9r2 ~ rdbms]$ cd
```

```
[oracle@oel7v9r2 ~]$ rm -r clone/
```

※ 원래 데이터베이스로 들어가려면 SID 를 다시 적용.

```
[oracle@oel7v9r2 ~]$ . oraenv
```

```
ORACLE_SID = [clone] ? orcl
```

<<시나리오 2>> RMAN 을 이용한 Oracle DB 복제 (Active Duplication)

1. 복제 DB 에 패스워드 파일 생성

```
$ orapwd file=orapwNEWDB format=12 force=y password=oracle
```

2. 소스 DB쪽 리스너에 SID정보 추가

```
$ vi listener.ora
```

```
SID_LIST_LISTENER =
```

```
(SID_LIST =
```

```
(SID_DESC =
```

```
(SID_NAME = NEWDB)
```

```
(ORACLE_HOME = /u01/app/oracle/product/19.3.0/dbhome_1)
```

```
)
```

```
)
```

3. 소스/ 복제 DB와 연결할 수 있는 TNS 설정

```
$ vi tnsnames.ora
```

```
T1 =
```

```
(DESCRIPTION =
```

```
(ADDRESS_LIST =
```

```
(ADDRESS = (PROTOCOL = TCP)(HOST = oel7v9r1.localdomain)(PORT = 1521))
```

```
)
```

```
(CONNECT_DATA =
```

```
(SERVICE_NAME = orcl.localdomain)
```

```
)
```

```
)
```

```
C1 =
```

```
(DESCRIPTION =
```

```
(ADDRESS_LIST =
```

```
(ADDRESS = (PROTOCOL = TCP)(HOST = prod.localdomain)(PORT = 1521))
```

```
)
```

```
(CONNECT_DATA =
```

```
(SERVICE_NAME = prod.localdomain)
```

```
)
```

```
)
```

```
LISTENER_ORCL =
```

```
(ADDRESS = (PROTOCOL = TCP)(HOST = oel7v9r1.localdomain)(PORT = 1521))
```

N1 =

```
(DESCRIPTION =  
  (ADDRESS_LIST =  
    (ADDRESS = (PROTOCOL = TCP)(HOST = prod.localdomain)(PORT = 1521))  
  )  
  (CONNECT_DATA =  
    (SERVICE_NAME = NEWDB.localdomain)  
  )  
)
```

4. 소스 DB 접속 확인

```
$ sqlplus sys@t1 as sysdba  
$ sqlplus sys@n1 as sysdba
```

5. 복제 DB 쪽에 필요한 디렉토리 생성

```
$ mkdir -p /u01/app/oracle/oradata/NEWDB  
$ mkdir -p /u01/app/oracle/oradata/NEWDB/adump  
$ mkdir -p /u01/app/oracle/oradata/NEWDB/fast_recovery_area
```

6. 파라미터 파일(initNEWDB.ora) 생성

```
$ vi $ORACLE_HOME/dbs/initNEWDB.ora  
db_name=NEWDB  
REMOTE_LOGIN_PASSWORDFILE=EXCLUSIVE
```

7. 복제 DB를 nomount .

```
$ export ORACLE_HOME=/u01/app/oracle/product/19.3.0/dbhome_1  
$ export ORACLE_SID=NEWDB  
$ sqlplus / as sysdba  
SQL> startup nomount pfile='$ORACLE_HOME/dbs/initNEWDB.ora'
```

8. 복제 DB생성하기

```
[oracle@oel7v9r1 admin]$ rman target "'sys@t1 as sysdba'" auxiliary "'sys@n1 as sysdba'"  
target database Password:  
connected to target database: ORCL (DBID=1727762462)  
auxiliary database Password:  
connected to auxiliary database: NEWDB (not mounted)
```

```
RMAN> duplicate target database to NEWDB from active database  
2> spfile  
3> set db_name='NEWDB'  
4> set control_files='/u01/app/oracle/oradata/NEWDB/control01.ctl'
```

```
5> set db_file_name_convert='/u01/app/oracle/oradata/ORCL', '/u01/app/oracle/oradata/NEWDB'
6> set log_file_name_convert='/u01/app/oracle/oradata/ORCL', '/u01/app/oracle/oradata/NEWDB'
7> set db_recovery_file_dest_size='1G'
8> set db_recovery_file_dest='/u01/app/oracle/oradata/NEWDB/fast_recovery_area'
9> set sga_target='900M'
10> set audit_file_dest='/u01/app/oracle/oradata/NEWDB/adump'
11> nofilenamecheck ;
```

Starting Duplicate Db at 20-MAY-25

using target database control file instead of recovery catalog

allocated channel: ORA_AUX_DISK_1

channel ORA_AUX_DISK_1: SID=1400 device type=DISK

current log archived

contents of Memory Script:

```
{
  restore clone from service 't1' spfile to
  '/u01/app/oracle/product/19.3.0/dbhome_1/dbs/spfileNEWDB.ora';
  sql clone "alter system set spfile= '/u01/app/oracle/product/19.3.0/dbhome_1/dbs/spfileNEWDB.ora'";
}
```

executing Memory Script

Starting restore at 20-MAY-25

using channel ORA_AUX_DISK_1

channel ORA_AUX_DISK_1: starting datafile backup set restore

channel ORA_AUX_DISK_1: using network backup set from service t1

channel ORA_AUX_DISK_1: restoring SPFILE

output file name=/u01/app/oracle/product/19.3.0/dbhome_1/dbs/spfileNEWDB.ora

channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:04

Finished restore at 20-MAY-25

sql statement: alter system set spfile= '/u01/app/oracle/product/19.3.0/dbhome_1/dbs/spfileNEWDB.ora'

contents of Memory Script:

```
{
  sql clone "alter system set db_name =
  'NEWDB' comment=
  'duplicate' scope=spfile";
  sql clone "alter system set db_name =
  'NEWDB' comment=
```

```
"" scope=spfile";
sql clone "alter system set control_files =
'/u01/app/oracle/oradata/NEWDB/control01.ctl'" comment=
"" scope=spfile";
sql clone "alter system set db_file_name_convert =
'/u01/app/oracle/oradata/ORCL', '/u01/app/oracle/oradata/NEWDB'" comment=
"" scope=spfile";
sql clone "alter system set log_file_name_convert =
'/u01/app/oracle/oradata/ORCL', '/u01/app/oracle/oradata/NEWDB'" comment=
"" scope=spfile";
sql clone "alter system set db_recovery_file_dest_size =
1G comment=
"" scope=spfile";
sql clone "alter system set db_recovery_file_dest =
'/u01/app/oracle/oradata/NEWDB/fast_recovery_area'" comment=
"" scope=spfile";
sql clone "alter system set sga_target =
900M comment=
"" scope=spfile";
sql clone "alter system set audit_file_dest =
'/u01/app/oracle/oradata/NEWDB/adump'" comment=
"" scope=spfile";
shutdown clone immediate;
startup clone nomount;
}
```

executing Memory Script

sql statement: alter system set db_name = "NEWDB" comment= "duplicate" scope=spfile

sql statement: alter system set db_name = "NEWDB" comment= "" scope=spfile

sql statement: alter system set control_files = '/u01/app/oracle/oradata/NEWDB/control01.ctl' comment= "" scope=spfile

sql statement: alter system set db_file_name_convert = '/u01/app/oracle/oradata/ORCL',
'/u01/app/oracle/oradata/NEWDB' comment= "" scope=spfile

sql statement: alter system set log_file_name_convert = '/u01/app/oracle/oradata/ORCL',
'/u01/app/oracle/oradata/NEWDB' comment= "" scope=spfile

sql statement: alter system set db_recovery_file_dest_size = 1G comment= "" scope=spfile

```
sql statement: alter system set db_recovery_file_dest = '/u01/app/oracle/oradata/NEWDB/fast_recovery_area'
comment= '' scope=spfile
```

```
sql statement: alter system set sga_target = 900M comment= '' scope=spfile
```

```
sql statement: alter system set audit_file_dest = '/u01/app/oracle/oradata/NEWDB/adump' comment= ''
scope=spfile
```

Oracle instance shut down

connected to auxiliary database (not started)

Oracle instance started

Total System Global Area 943716976 bytes

Fixed Size 8903280 bytes

Variable Size 301989888 bytes

Database Buffers 624951296 bytes

Redo Buffers 7872512 bytes

contents of Memory Script:

```
{
  sql clone "alter system set db_name =
'ORCL' comment=
'Modified by RMAN duplicate' scope=spfile";
  sql clone "alter system set db_unique_name =
'NEWDB' comment=
'Modified by RMAN duplicate' scope=spfile";
  shutdown clone immediate;
  startup clone force nomount
  restore clone from service 't1' primary controlfile;
  alter clone database mount;
}
```

executing Memory Script

```
sql statement: alter system set db_name = 'ORCL' comment= 'Modified by RMAN duplicate' scope=spfile
```

```
sql statement: alter system set db_unique_name = 'NEWDB' comment= 'Modified by RMAN duplicate'
scope=spfile
```

Oracle instance shut down

Oracle instance started

Total System Global Area 943716976 bytes

Fixed Size 8903280 bytes

Variable Size 301989888 bytes

Database Buffers 624951296 bytes

Redo Buffers 7872512 bytes

Starting restore at 20-MAY-25

allocated channel: ORA_AUX_DISK_1

channel ORA_AUX_DISK_1: SID=247 device type=DISK

channel ORA_AUX_DISK_1: starting datafile backup set restore

channel ORA_AUX_DISK_1: using network backup set from service t1

channel ORA_AUX_DISK_1: restoring control file

channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:12

output file name=/u01/app/oracle/oradata/NEWDB/control01.ctl

Finished restore at 20-MAY-25

database mounted

contents of Memory Script:

```
{
  set newname for datafile 1 to
"/u01/app/oracle/oradata/NEWDB/system01.dbf";
  set newname for datafile 3 to
"/u01/app/oracle/oradata/NEWDB/sysaux01.dbf";
  set newname for datafile 4 to
"/u01/app/oracle/oradata/NEWDB/undotbs01.dbf";
  set newname for datafile 5 to
"/u01/app/oracle/oradata/NEWDB/ts01.dbf";
  set newname for datafile 7 to
"/u01/app/oracle/oradata/NEWDB/users01.dbf";
  restore
  from nonsparse from service
't1' clone database
;
sql 'alter system archive log current';
```

}

executing Memory Script

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executingcommand: SET NEWNAME

Starting restore at 20-MAY-25

using channel ORA_AUX_DISK_1

channel ORA_AUX_DISK_1: starting datafile backup set restore

channel ORA_AUX_DISK_1: using network backup set from service t1

channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_AUX_DISK_1: restoring datafile 00001 to /u01/app/oracle/oradata/NEWDB/system01.dbf

channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:04:10

channel ORA_AUX_DISK_1: starting datafile backup set restore

channel ORA_AUX_DISK_1: using network backup set from service t1

channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_AUX_DISK_1: restoring datafile 00003 to /u01/app/oracle/oradata/NEWDB/sysaux01.dbf

channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:01:38

channel ORA_AUX_DISK_1: starting datafile backup set restore

channel ORA_AUX_DISK_1: using network backup set from service t1

channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_AUX_DISK_1: restoring datafile 00004 to /u01/app/oracle/oradata/NEWDB/undotbs01.dbf

channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:26

channel ORA_AUX_DISK_1: starting datafile backup set restore

channel ORA_AUX_DISK_1: using network backup set from service t1

channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_AUX_DISK_1: restoring datafile 00005 to /u01/app/oracle/oradata/NEWDB/ts01.dbf

channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:01

channel ORA_AUX_DISK_1: starting datafile backup set restore

channel ORA_AUX_DISK_1: using network backup set from service t1

channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_AUX_DISK_1: restoring datafile 00007 to /u01/app/oracle/oradata/NEWDB/users01.dbf

channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:01:19

Finished restore at 20-MAY-25

sql statement: alter system archive log current

current log archived

contents of Memory Script:

```
{
    restore clone force from service 't1'
        archivelog from scn 3185489;
    switch clone datafile all;
}
```

executing Memory Script

Starting restore at 20-MAY-25

using channel ORA_AUX_DISK_1

channel ORA_AUX_DISK_1: starting archived log restore to default destination

channel ORA_AUX_DISK_1: using network backup set from service t1

channel ORA_AUX_DISK_1: restoring archived log

archived log thread=1 sequence=98

channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:02

channel ORA_AUX_DISK_1: starting archived log restore to default destination

channel ORA_AUX_DISK_1: using network backup set from service t1

channel ORA_AUX_DISK_1: restoring archived log

archived log thread=1 sequence=99

channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:01

Finished restore at 20-MAY-25

datafile 5 switched to datafile copy

input datafile copy RECID=12 STAMP=1201627636 file name=/u01/app/oracle/oradata/NEWDB/ts01.dbf

datafile 1 switched to datafile copy

input datafile copy RECID=13 STAMP=1201627638 file name=/u01/app/oracle/oradata/NEWDB/system01.dbf

datafile 3 switched to datafile copy

input datafile copy RECID=14 STAMP=1201627638 file name=/u01/app/oracle/oradata/NEWDB/sysaux01.dbf

datafile 4 switched to datafile copy

input datafile copy RECID=15 STAMP=1201627638 file name=/u01/app/oracle/oradata/NEWDB/undotbs01.dbf

datafile 7 switched to datafile copy

input datafile copy RECID=16 STAMP=1201627638 file name=/u01/app/oracle/oradata/NEWDB/users01.dbf

contents of Memory Script:

```
{
    set until scn 3185861;
    recover
    clone database
    delete archivelog
    ;
}
```

```
}
```

executing Memory Script

executing command: SET until clause

Starting recover at 20-MAY-25

using channel ORA_AUX_DISK_1

datafile 5 not processed because file is read-only

starting media recovery

archived log for thread 1 with sequence 98 is already on disk as file

/home/oracle/arch20001_0000000098_1200585133.arc

archived log for thread 1 with sequence 99 is already on disk as file

/home/oracle/arch20001_0000000099_1200585133.arc

archived log file name=/home/oracle/arch20001_0000000098_1200585133.arc thread=1 sequence=98

archived log file name=/home/oracle/arch20001_0000000099_1200585133.arc thread=1 sequence=99

media recovery complete, elapsed time: 00:00:02

Finished recover at 20-MAY-25

contents of Memory Script:

```
{
```

```
    delete clone force archivelog all;
```

```
}
```

executing Memory Script

released channel: ORA_AUX_DISK_1

allocated channel: ORA_DISK_1

channel ORA_DISK_1: SID=16 device type=DISK

deleted archived log

archived log file name=/home/oracle/arch20001_0000000098_1200585133.arc RECID=1 STAMP=1201627633

deleted archived log

archived log file name=/home/oracle/arch20001_0000000099_1200585133.arc RECID=2 STAMP=1201627634

Deleted 2 objects

Oracle instance started

Total System Global Area 943716976 bytes

Fixed Size 8903280 bytes

Variable Size 301989888 bytes

Database Buffers 624951296 bytes
Redo Buffers 7872512 bytes

contents of Memory Script:

```
{  
    sql clone "alter system set  db_name =  
    'NEWDB' comment=  
    'Reset to original value by RMAN' scope=spfile";  
    sql clone "alter system reset  db_unique_name scope=spfile";  
}
```

executing Memory Script

sql statement: alter system set db_name = "NEWDB" comment= "Reset to original value by RMAN"
scope=spfile

sql statement: alter system reset db_unique_name scope=spfile
Oracle instance started

Total System Global Area 943716976 bytes

Fixed Size 8903280 bytes
Variable Size 301989888 bytes
Database Buffers 624951296 bytes
Redo Buffers 7872512 bytes

sql statement: CREATE CONTROLFILE REUSE SET DATABASE "NEWDB" RESETLOGS ARCHIVELOG

```
MAXLOGFILES        16  
MAXLOGMEMBERS       3  
MAXDATAFILES       100  
MAXINSTANCES       8  
MAXLOGHISTORY       292
```

LOGFILE

```
GROUP        1 ( '/u01/app/oracle/oradata/NEWDB/redo01.log' ) SIZE 10 M REUSE,  
GROUP        2 ( '/u01/app/oracle/oradata/NEWDB/redo02.log' ) SIZE 10 M REUSE,  
GROUP        3 ( '/u01/app/oracle/oradata/NEWDB/redo03.log' ) SIZE 10 M REUSE
```

DATAFILE

```
'/u01/app/oracle/oradata/NEWDB/system01.dbf'
```

CHARACTER SET AL32UTF8

contents of Memory Script:

```
{  
    set newname for tempfile 1 to  
    "/u01/app/oracle/oradata/NEWDB/temp01.dbf";
```

```
switch clone tempfile all;
catalog clone datafilecopy  "/u01/app/oracle/oradata/NEWDB/sysaux01.dbf",
"/u01/app/oracle/oradata/NEWDB/undotbs01.dbf",
"/u01/app/oracle/oradata/NEWDB/users01.dbf";
switch clone datafile all;
}
```

executing Memory Script

executing command: SET NEWNAME

renamed tempfile 1 to /u01/app/oracle/oradata/NEWDB/temp01.dbf in control file

cataloged datafile copy

datafile copy file name=/u01/app/oracle/oradata/NEWDB/sysaux01.dbf RECID=1 STAMP=1201627692

cataloged datafile copy

datafile copy file name=/u01/app/oracle/oradata/NEWDB/undotbs01.dbf RECID=2 STAMP=1201627692

cataloged datafile copy

datafile copy file name=/u01/app/oracle/oradata/NEWDB/users01.dbf RECID=3 STAMP=1201627692

datafile 3 switched to datafile copy

input datafile copy RECID=1 STAMP=1201627692 file name=/u01/app/oracle/oradata/NEWDB/sysaux01.dbf

datafile 4 switched to datafile copy

input datafile copy RECID=2 STAMP=1201627692 file name=/u01/app/oracle/oradata/NEWDB/undotbs01.dbf

datafile 7 switched to datafile copy

input datafile copy RECID=3 STAMP=1201627692 file name=/u01/app/oracle/oradata/NEWDB/users01.dbf

contents of Memory Script:

```
{
  Alter clone database open resetlogs;
}
```

executing Memory Script

database opened

contents of Memory Script:

```
{
  catalog clone datafilecopy  "/u01/app/oracle/oradata/NEWDB/ts01.dbf";
  switch clone datafile  5 to datafilecopy
"/u01/app/oracle/oradata/NEWDB/ts01.dbf";
  #online the readonly tablespace
  sql clone "alter tablespace  TS1 online";
}
```

executing Memory Script

cataloged datafile copy

datafile copy file name=/u01/app/oracle/oradata/NEWDB/ts01.dbf RECID=4 STAMP=1201627766

datafile 5 switched to datafile copy

input datafile copy RECID=4 STAMP=1201627766 file name=/u01/app/oracle/oradata/NEWDB/ts01.dbf

sql statement: alter tablespace TS1 online

Finished Duplicate Db at 20-MAY-25

9. 복제 DB 확인하기

```
[oracle@prod dbs]$ sqlplus / as sysdba
```

Connected to:

Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production

Version 19.3.0.0.0

```
SQL> select instance_name from v$instance;
```

INSTANCE_NAME

NEWDB

```
SQL> select status from v$instance;
```

STATUS

OPEN

```
SQL> select name from v$datafile union
```

```
2 select name from v$controlfile union
```

```
3 select name from v$tempfile union
```

```
4 select member from v$logfile;
```

NAME

/u01/app/oracle/oradata/NEWDB/control01.ctl

/u01/app/oracle/oradata/NEWDB/redo01.log

/u01/app/oracle/oradata/NEWDB/redo02.log

/u01/app/oracle/oradata/NEWDB/redo03.log

/u01/app/oracle/oradata/NEWDB/sysaux01.dbf

/u01/app/oracle/oradata/NEWDB/system01.dbf

/u01/app/oracle/oradata/NEWDB/temp01.dbf

/u01/app/oracle/oradata/NEWDB/ts01.dbf

/u01/app/oracle/oradata/NEWDB/undotbs01.dbf

/u01/app/oracle/oradata/NEWDB/users01.dbf

10 rows selected.